

Math 2270 Review for Midterm 1

Ex Show that the set of invertible matrices (resp. singular matrices) is not a subspace of M § 1.1-3.5

Ex Show that the set of symmetric matrices forms a subspace of M

Ex Show that $\cos x, \sin x$ are LI, $1, x, x^2$ are LI but $\cos^2 x, \sin^2 x$ are LD (LI = lin independent, LD = lin dependent)

Ex For which b_1, b_2, b_3 can you solve:

a) $\begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ b) $\begin{bmatrix} 1 & 4 \\ 2 & 9 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ c) $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

describe all possible solutions.

Ex Compute the null spaces of $\begin{bmatrix} 1 & 2 & 0 & 3 & 5 \\ 1 & 2 & 1 & 5 & 7 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 4 \\ 0 & 2 & -1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 2 & 2 & 10 \\ 1 & 1 & 5 \end{bmatrix}$

Ex Compute all sols of $\begin{bmatrix} 1 & 2 & 0 & 3 & 5 \\ 1 & 2 & 1 & 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ u \\ v \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$

x Compute the rank and nullspace of $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$ total $\in \mathbb{R}$

x Find a matrix with nullspace $\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rangle$ (resp. $\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rangle$)

x Find a system of equations with solutions $\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ (resp. $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + s \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$)

Ex Explain why 4 vectors in $\mathbb{R}^3$ can not be LI.

x Explain " 2 vectors in $\mathbb{R}^3$ can not generate $\mathbb{R}^3$.

Ex Prove that $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ are LD, $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ are LI.